

Predicting Hospital Readmissions for Diabetic Patients Using Machine Learning

Abstract

Readmissions to hospitals continue to be a major problem in healthcare which could be due to various diseases, inadequate treatment, medical errors, early discharge, and other socioeconomic factors. Identifying patients at a high risk of readmission help hospitals allocate resources efficiently and reduce unnecessary readmissions, which further cut down the costs. In today's world, Diabetes, a chronic condition, has become very common across all age groups, which accounts for higher readmission rates due to complications and poor management. This project is to predict hospital readmissions for diabetic patients using a large dataset that includes demographic data, medical history and other clinical features, focusing on whether factors like glucose levels, A1C results, and medication usage are key predictors of readmission. Various machine learning models were used to predict the readmissions. The models are then evaluated using scoring metrics like accuracy, recall, F1score and AUC-ROC. Class imbalance was addressed using balanced class weights and SMOTE.

1. Introduction

Hospital readmissions have been a major concern for the healthcare administrators, policymakers and clinicians from the past few decades. The readmission of a patient within 30 days is very much alarming and impactful to the healthcare providers, as it requires a large amount of resources to be allocated to the patients which could be the medicines, multiple diagnoses, and needs of equipment related to laboratories or bed allocation which is very expensive. This project is particular focused on Diabetes patients as it is a chronic disease that is being faced by small and young

children today. This could be caused due to various factors like obesity, food intake, alcohol consumption, family history, other health conditions, and many more. Also, diabetes is one of the conditions which requires frequent monitory to check the levels of glucose if there's a spike or if the condition is stable. Identifying patients with high risk of readmission can enable early intervention and help prevent risk to the patient's life.

1.1.1 Overview of the Data

This study uses the data from Kaggle, which has over 100,000 diabetic patient encounters from 130-US hospitals from 1999-2008. The dataset has 50 features (13 numerical and 37 categorical) focusing on the demographics, admission details, medical history, lab test results, medications and hospital stay information. Some of the key features include admission type, number of medications, discharge disposition, length of stay, A1C test results, number of procedures and total visits. The target variable 'readmission' focuses on three classes <30, >30 and NO indicating readmissions within 30 days after 30 days and no readmissions respectively. The dataset contains a lot of missing information for the weight, payer_code, medical speciality, race and diagnoses. The dataset is highly imbalanced with about 11% of patients being readmitted within 30 days of discharge. Missing values are imputed, numerical features are normalized whilst handling collinearity and outliers. Also, for this study the target variable is transformed into a binary classification problem by combining >30 and No together so the main focus relies on predicting the readmissions within 30 days.

2. Literature Review

There has been an extensive research on the readmission predictions for diabetes patients. Lama

Alturki [1] identifies the predictors for readmission as well as the length of stay as the long stays at hospitals requires huge amount of resources and high costs are involved. The study [2] identifies that number of admissions, age, diagnosis were the main characteristics used to identify the probability of accidental readmission which is the same with my study as mentioned in section 3.1. The study also aligns with Yujuan Shang's take on identifying the more admission times were the main group readmitted within 30 days and were mostly elderly. The diagnosis data is grouped to smaller groups using standard ICD9 codes [3] [4] [5] which has reduced a lot of data. Various machine learning models were used to predict the readmissions for diabetic patients in the studies. Omar Hasan in [6] uses only the logistic regression model and achieved an AUC-Score of 0.64 which our study is perfectly aligned. It solely relies on one linear model which does not capture complex relationships and is not a good prediction model. The study [5] explores into the deep learning models and achieved an improvement in the scoring metrics compared to the convolution neural networks (CNN). Rene Rodriguez [4] identifies in his study the racial and ethnic differences in 30-Day hospital readmissions among US adults with diabetes. He found that the white patients have higher readmissions while our study using data from various hospitals found Caucasians with high number of readmissions. The resampling techniques were used to address class imbalance where undersampling performed better than the oversampling likely due to the synthetic generated data. The study [1] found the SVM and KNN as the best models which our study doesn't explore. XGBoost is the best performing model in our study for predicting the readmissions with a recall of 0.42 and AUC score of 0.66 which fairly balances the tradeoff. Future studies should include SVM and KNN to draw conclusions on better performing models.

3. Methodology

3.1 Exploratory Data Analysis

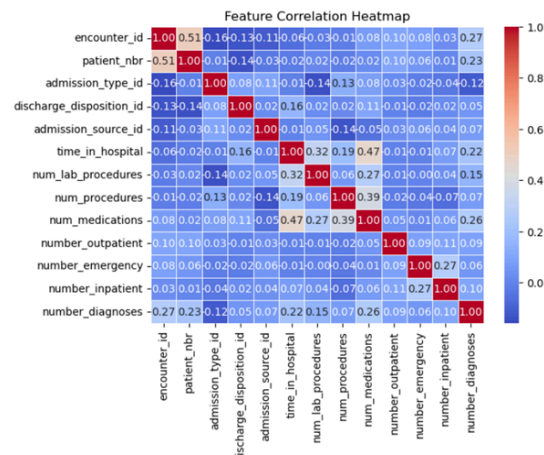
Various plots like histograms, boxplots and barplots were used to identify the outliers, missing values and understand feature distributions. Plots were

also used to understand the importance of features on the distributions which revealed insightful patterns.

Observations:

- People of ages between 70 and 80 are likely to face readmissions. This is valid because diabetes tend to be higher in elder population.
- The higher the number of medications prescribed, higher the chance of readmission. It implies how critical the disease as more medications were being used.
- The medicine dosage prescribed also shows an affect as a change in the dosage from stable to high means there is a chance of the patient visiting the hospital.
- It is also observed that the 'Caucasian' people are high in readmissions.

Multicollinearity is also identified between the numerical features using the Pearson Correlation matrix which helps identify the highly correlated features which might affect the predictive power of the models.



3.2 Data Preprocessing

Upon checking all the features, it is observed that the 'diabetesMed' tracks if the diabetes medicine was being used, 'change' identified if there's any change in the medication (either the dosage or the medicine itself). Considering these features tracking most of the information related to the medications, all the 24 individual medication features were dropped.

3.2.1 Handling Null Values

The missing values for weight (97%), payer_code (49%) and medical_speciality(39%) were very high, and hence were dropped from the dataset. Also, the unique identifiers patient_nbr and encounter_id were removed since they don't contribute in predicting the readmissions. The values of 'race', 'max_glu_serum', 'A1Cresult' were imputed properly with 'Unknown' and 'None' (no result) and a few missing values of the diagnoses records were removed. Also, the duplicate records were checked and dropped.

3.2.2 Log Transformations

The inpatient, outpatient, emergency visits and number of medications are skewed. The log transformations are applied to these variables to normalize them before training the models.

3.2.3 Variable Encoding

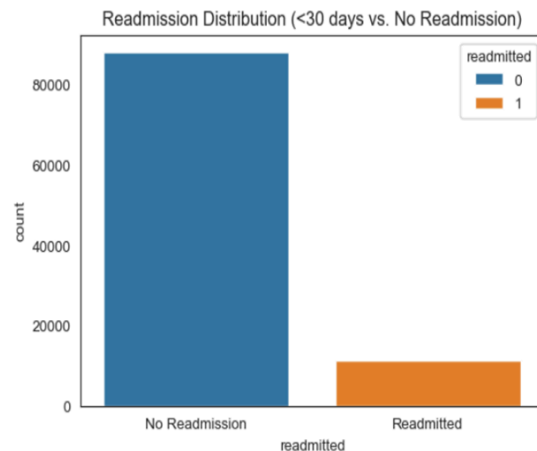
The variables admission_type, admission source and discharge_disposition_id are categorical but are represented as numbers in the data. The corresponding mapping data was provided in an ID_Mappings file. The discharge_disposition_id corresponds to 29 categories such as discharged to home, hospice, expired and not available. The admission_source corresponds to 21 categories like physician referral, emergency room and transfer from a hospital. The admission_type indicates if it was an emergency, urgent or newborn. Due to the high cardinality of the features, they are label encoded to multiple values and removed data of expired patients. The 'gender', 'change' and 'diabetesMed' are binary encoded while 'race', 'max_glu_serum', 'A1Cresult' were one hot encoded.

3.2.4 Feature Engineering

The three diagnoses related feature 'diag_1', 'diag_2' and 'diag_3' have thousands of values which were mapped into nine broader disease categories(0-8) by using the ICD9 codes. The mapped disease categories are: Circulatory, Respiratory, Digestive, Diabetes, Injury, Musculoskeletal, Genitourinary, Neoplasms and Other/Unclassified(0). This helps identify patients grouped by particular diagnosis. A new feature

'num_visits' was created by aggregating the number of inpatient, outpatient and emergency visits. This gives an overall combined view instead of individual contributions.

3.3 Addressing Class Imbalance



Since only 11% of patients were readmitted, class imbalance posed a significant challenge. Various techniques like SMOTE (OverSampling) to generate synthetic samples for the minority class, Class weighting in Logistic Regression, Decision Trees, and Random Forest, and Balanced Bagging to counteract bias and improve recall. There is a drastic improvement in the models' behavior with class imbalance handling. Without handling it, the logistic regression model achieved 89% accuracy with no 0% of recall and precision. Upon using the balanced class weight, it jumped to 56% recall and 17% precision while the accuracy has dropped which is fine since the model aims to predict readmissions correctly which minimizes the risks associated. The SMOTE was performing better than the balanced classweight which performed very poor with recall close to 0.15.

3.4 Feature Selection

The feature selection was performed using the Random Forest RFE method to identify and select the top performing features contributing to the prediction of the hospital readmissions. The selected features are: 'age', 'admission_type_id', 'discharge_disposition_id', 'admission_source_id', 'time_in_hospital', 'race', 'num_lab_procedures', 'number_diagnoses', 'change', 'diabetesMed', 'ICD9_code', 'max_glu_serum_None', 'A1Cresult_>8',

'A1Cresult_None', 'num_medications_log' and 'num_visits_log'.

3.5 Model Training and Evaluation

The models used in this study are Logistic Regression, Decision Trees, Random Forest, XGBoost, Bagging and Ensemble with Voting Classifier. All models are trained on 80% of the data and is tested on the other 20% of the data. The random_state - 42 was used to maintain consistency across all the results.

Evaluation Metrics: Accuracy for overall model's performance, Precision and Recall to trade-off between reducing false positives and capturing actual readmissions, F1-score to balance between precision and recall and AUC-ROC score to measure model's discrimination ability.

Logistic Regression served as a baseline model and performed the worst among all models. Feature scaling was done to ensure standardization of the variables. The model could not capture the complex relationships between features due to its linear nature. It achieved a recall of only 0.57, and it failed to identify high-risk patients. Despite handling imbalances using class weights, the model show much improvement in predictive performance. Recall is very important in a healthcare dataset with many dependencies. The best model resulted after creating model with C=0.01 and using 'liblinear' solver.

Decision Tree model has achieved 81% accuracy with 15% recall which is poor at predicting the readmissions correctly. The tuned decision tree model improved to a higher recall of 0.64 after optimizing max_depth:5, min_samples_leaf:5, min_samples_split:2, but the model achieved only 57% accuracy. This indicates its nature to overfitting. It clear struggles to distinguish between the models and the performance is not good.

Random Forest model achieved an AUC-ROC of 0.6493, indicating strong predictive capability. However, it suffered from class imbalance, predicting majority-class predictions. Applying SMOTE increased recall but resulted in lower precision (0.18) with only 0.14 recall, indicating that it predicts more false positives. Although this trade-off has a little balance, it is not the ideal

model. The tuned model has shown an improvement in the recall with 55% but the accuracy dropped from 83% to 66% with parameters 'max_depth':10, 'min_samples_split':10, and 'n_estimators': 300

XGBoost outperformed all individual models with an AUC-ROC of 0.66, indicating it was the most effective in properly classifying the readmitted and non-readmitted patients. Adjusting n_estimators, max_depth, learning_rate, and scale_pos_weight, recall was increased to 0.42, allowing more high-risk patients to be identified. It also achieves the highest precision of 0.20 and accuracy as well outperformed with 0.74 maintaining the best trade off balance. The optimal parameters were: 'subsample': 0.7, 'n_estimators': 300, 'max_depth': 5, 'learning_rate': 0.01, 'colsample_bytree': 0.7.

Balanced Bagging (Bootstrap Aggregation) model was used to balance the bias tradeoff addressing the class imbalance problem in the data and give robust predictions. This model has fairly performed better than the baseline models with the highest recall of 0.58 means it correctly classifies 58% of the readmissions. While having an accuracy of 65%, it fairly balanced the precision at 0.18.

The ensemble model (Random Forest + XGBoost + Decision Tree) achieved an accuracy of 0.78, a recall of 0.35, and an AUC-ROC of 0.66, making it the best balanced model in the study. While XGBoost alone had the best recall, combining it with Random Forest and Decision Trees resulted in a more stable model with improved precision. The ensemble approach helped mitigate individual model weaknesses, ensuring both generalization and reliability. Ensemble was also trained using different combinations but the three model combination has resulted in the best performance.

4. Results

The predictive performance of various machine learning models was evaluated using key metrics, including accuracy, precision, recall, F1-score, and AUC-ROC. Since the dataset was highly imbalanced, recall and AUC-ROC were prioritized over accuracy to ensure the model could effectively identify high-risk patients who were likely to be

readmitted within 30 days. The models were trained and tested using (80-20) train-test splits, with hyperparameter tuning applied using GridSearchCV where applicable. Cross validation (Stratified K-Fold with 5 splits) was used to ensure robustness and avoid model overfitting.

4.1 Model Performance Comparison

The table summarizes the performance of all trained models, including the final best versions obtained after tuning.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0.63	0.17	0.56	0.26	0.64
Decision Trees	0.57	0.16	0.64	0.25	0.64
Random Forests	0.66	0.18	0.55	0.27	0.65
Gradient Boosting	0.74	0.20	0.42	0.27	0.66
Ensemble (Voting)	0.73	0.20	0.43	0.27	0.66
Balanced Bagging	0.65	0.18	0.58	0.27	0.66

In our data, it is very important to identify the patients who are at high-risk of readmissions which would mitigate the risk of patients' lives. Hence, in this case recall is very much important than the accuracy of the model. From the results, we can see that high recall was achieved by decision trees but the accuracy and precision is low. This is due to the overfitting nature of the decision tree models. Considering all the scoring metrics, XGBoost had shown the high AUC score of 0.6646 and a recall of 0.42 while the precision is really good and makes a difference from other models, making it the best individual model for identifying high-risk readmitted patients. However, the ensemble model (RF + XGB + DT) provided a better trade-off by improving recall slightly while maintaining the precision and AUC, making it the best overall model.

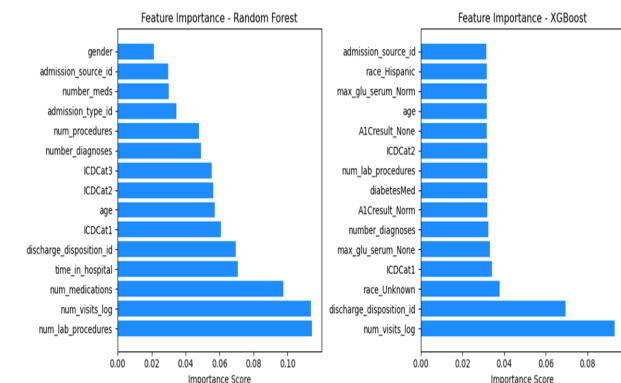
4.2 Five-fold Cross Validation

A five fold stratified cross validation was performed on all the test-train splits across all the models to find potential overfitting and found the results to be consistent and valid which confirms the models has achieved the best parameters found using GridSearchCV and is not overfitting.

4.3 Feature Importance Analysis

Using Random Forest and XGBoost feature importance metrics, the most important factors in predicting hospital readmissions were identified. The top contributing features included:

ICD-9 Diagnosis Category indicating medical conditions has an effect on increased readmission risks, **num_medications** indicate patients on multiple medications due to various diseases or issues, leading to readmission, **time_in_hospital** specifying the longer hospital stays correlated with higher readmission rates, **A1Cresult** indicates test results of glucose control which acts as a strong indicator of readmission, **number_visits** which says that patients with frequent hospital interactions were more likely to be readmitted and **discharge_disposition_id** indicating the affect of poor and inadequate care post discharge.



5. Discussion

The achieved results present both the benefits and challenges of using ML algorithms for identifying the diabetic hospital readmission. The findings align with research conducted by Shang et al., 2021, concluding that XGBoost and ensemble models perform better than others. However, problems such as model generalization, feature interactions and class imbalance were very challenging in predicting the outcomes. Due to the class dominance, models were skewed towards predicting patients with no readmissions as readmitted. SMOTE oversampling has increased false positives by decreasing precision while increasing recall. XGBoost captured complex relationships in patient data better than the Logistic Regression and Decision Trees. The ensemble

model provided a good tradeoff balance between recall and precision, ensuring fewer false positives while still identifying high-risk patients with a decent AUC Score.

These findings have certain implications, as hospitals can use such models in their systems to find patients at high-risk, monitor care taken post-discharges, change in the medication or the dosage, diagnosis results in order to reduce readmission rates. By using these predictive insights, health administrators can ensure timely follow-ups, medication adjustments, diagnosis, and monitoring at-risk patients. Alongside, the ethical considerations must also be addressed as the bias in training data can lead to misclassifications. AI-driven predictions should be transparent maintaining data privacy and security in compliance with healthcare regulations. Hospitals must use these AI/ML based models as decision-support tools rather than relying on them completely. Healthcare always needs a clinical intervention or advice from the healthcare providers and their knowledge is needed in decision-making by closely going through the patient's conditions.

Future work should explore models like SVM and KNN, incorporate real-world monitoring data, and refine predictions using important medication indicators like insulin and metformin use. Improving interpretability and balancing the bias is a key to ensuring a better model deployment in clinical settings.

6. Conclusion & Future Work

This study demonstrated how well machine learning algorithms can be used to predict hospital readmissions for diabetic patients. The findings suggest that XGBoost and ensemble models provide the best predictive capability, balancing recall and precision to ensure high-risk patients are identified while maintaining the precision. The disease categories, length of stay, number of visits, medication changes, and after-discharge care were found to be the most important features contributing to the prediction of readmission rates. Addressing class imbalance was a challenging issue due to highly imbalanced dataset. Applying SMOTE and class weights to the models helped improve the model's ability to identify readmitted patients. The final ensemble model using a voting

classifier was the most effective in categorizing the readmissions of the patients properly achieving fair AUC score and balancing model performance. This model can be integrated into hospital decision-making systems to help healthcare professionals proactively identify the high-risk patients before readmission occurs which would lead to better patient care and also improved health services while minimizing the costs.

Future improvements must focus on expanding the feature set of the data used to find out the predictive power of individual medications on the readmissions and performing PCA to reduce high-dimensionality. Also employing the advanced ML deep learning models would give great results and further help in the deployment of the models in real world to help healthcare providers by efficiently allocating the resources and save more on the medical costs. Further studies can enhance predictive accuracy and make hospital readmission prevention more effective, ultimately leading to better patient care and reduced healthcare costs.

7. References

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